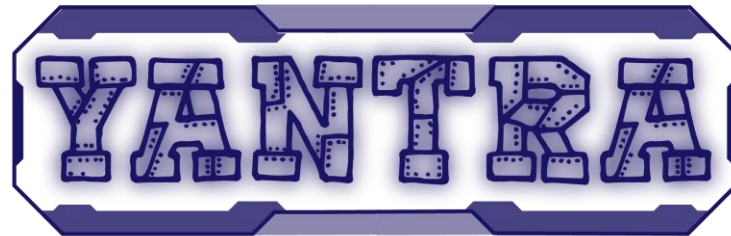


Software for RP



List of software for rapid Prototyping

1. Solid View
2. Magics
3. Mimics-Z
4. Magic Communicator

3. Mimics-z

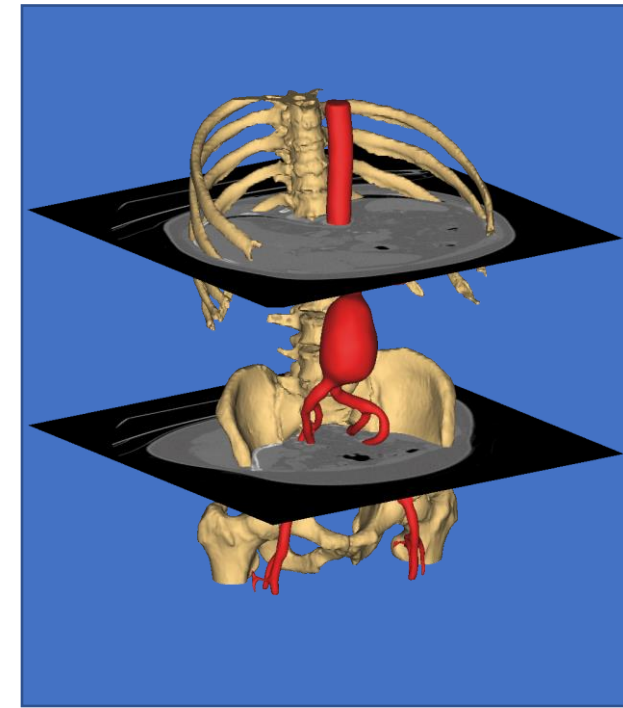
- For the first time, doctors, nurses, and technicians who have no previous experience with 3D modeling or 3D printing can create 3D anatomical models from MRI and CT scan images quickly and easily.
- Developed by **Materialise** (creator of Mimics, the leading medical imaging software for the rapid prototyping industry).
- Mimics Z™ is optimized for output on 3D Systems high-definition, 3D printers that produce **full-colour models** in few hours.
- Healthcare organizations worldwide increasingly rely on 3D anatomical models for **pre-operative planning, specialist consultation, implant fit and design, patient counselling and medical education.**



Magnetic resonance imaging (MRI)

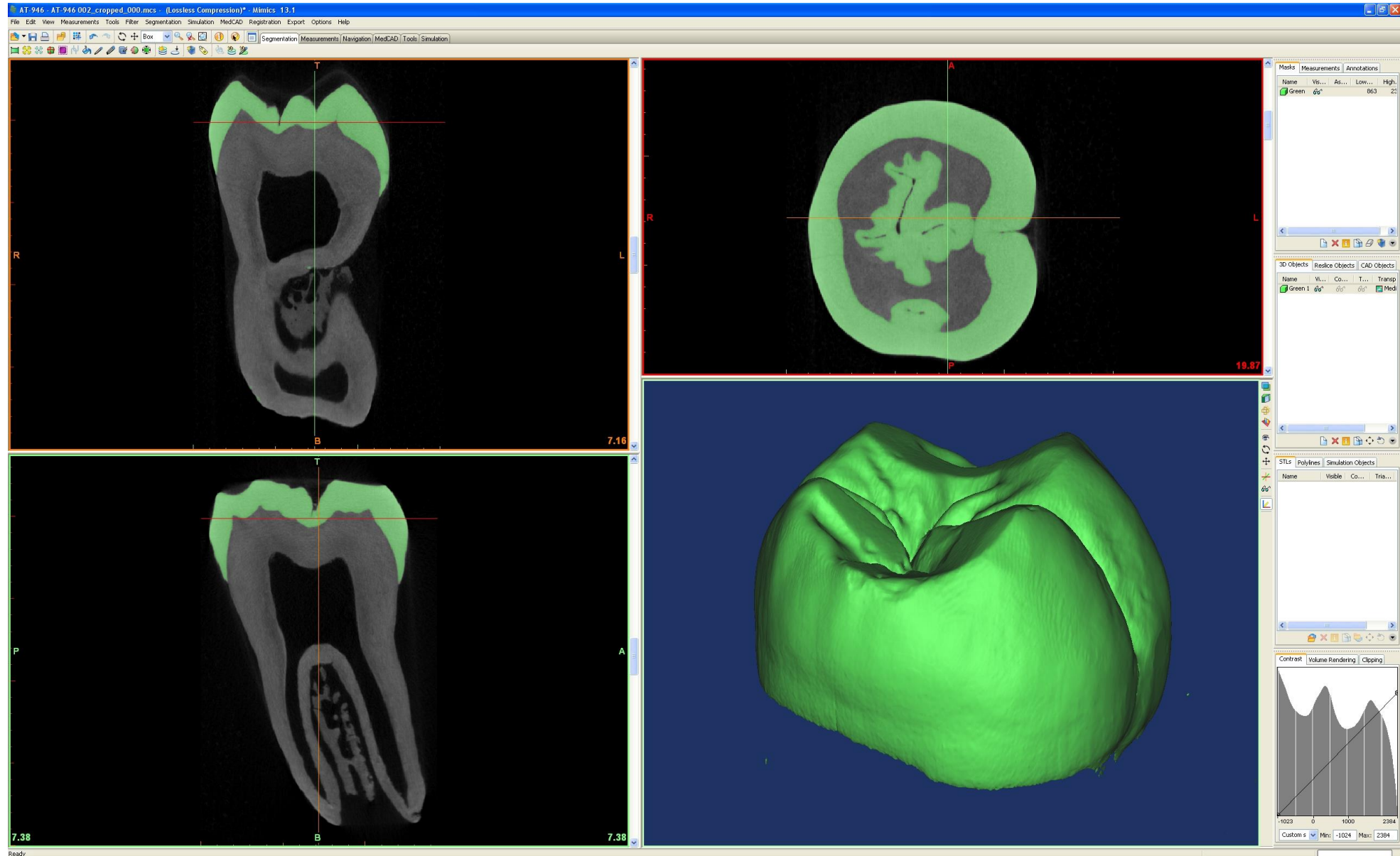
computerized tomography (CT)

- Bridge the gap between 2D image data & 3D engineering applications
- Mimics software allows you to process and **edit your 2D image** data (CT, iCT, MRI, etc.) to construct 3D models with the utmost accuracy, flexibility and user- friendliness.
- The powerful segmentation tools allow you to segment your medical CT/MRI images, **take measurements and engineer** directly on your 3D model.
- From there you can export your 3D data to a wide range of output formats and engineering applications; such as **FEA, design, surgical simulation, additive manufacturing and more.**
- Mimics is part of Materialise' total solution for advancing biomedical R&D



Magnetic resonance imaging (**MRI**)
computerized tomography (**CT**)

mimics-z software environment



Key features of mimics-z software

1. Software "wizards" **guide users** with **nominal training** through the entire process. Includes an extensive help function and templates.
2. **Automates all steps needed to import MRI and CT scan files**, select structures, conduct editing and masking, and export data.
3. Processes industry standard DICOM data and outputs ZPR files optimized for printing on 3D Systems ZPrinters.
4. Tight integration with ZEdit software, enabling users to add colour to highlight areas of interest (such as a bone tumour), and annotation features that make it possible to label models with critical patient data and doctor comment.

SOLID VIEW

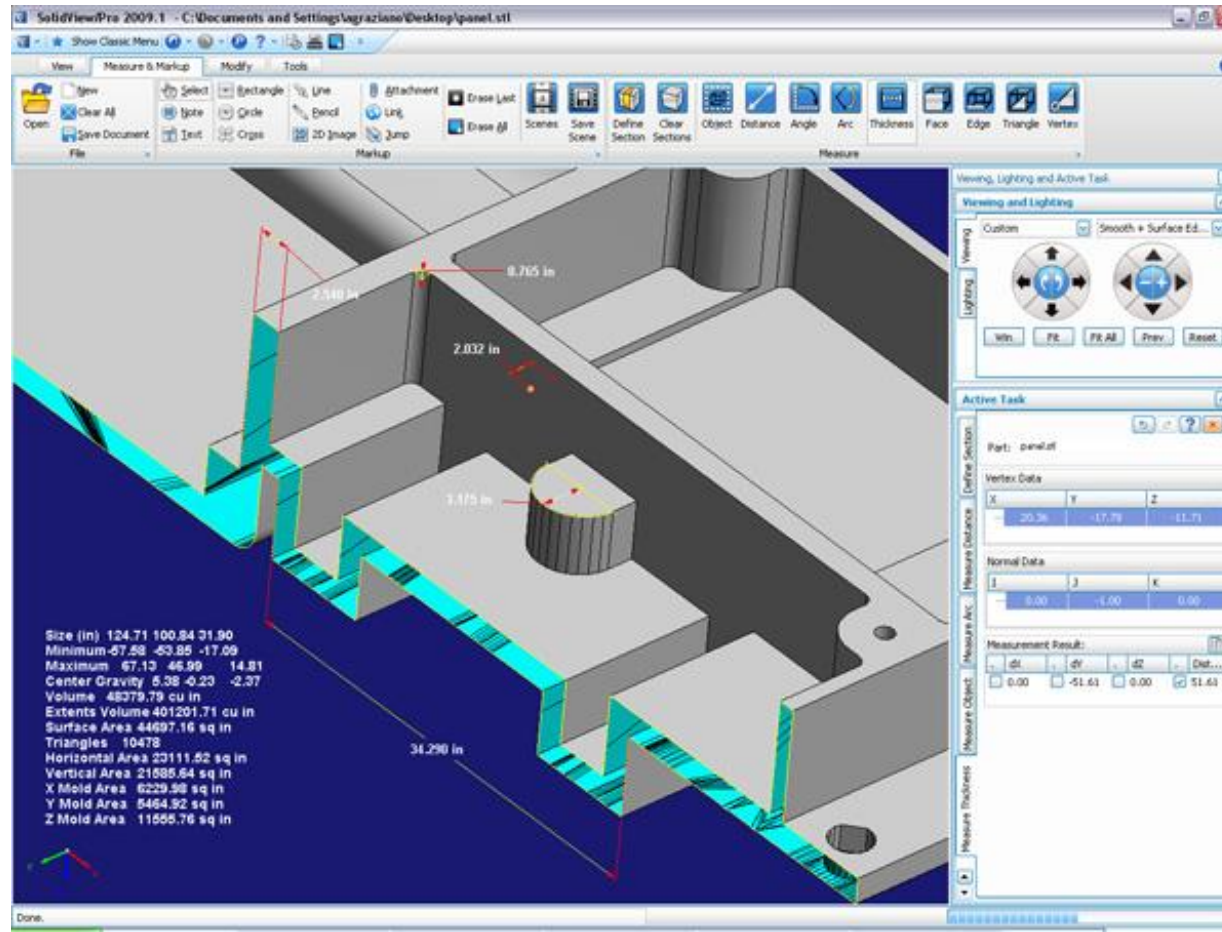
- Solid View software allows non-CAD users to
 1. Easily **view, measure, translate** and mark-up CAD data.
 2. Open up **communication** to all who need to be involved in the design process.
- Solid View is used across the world by those needing access to CAD data but **not trained** in using CAD systems.



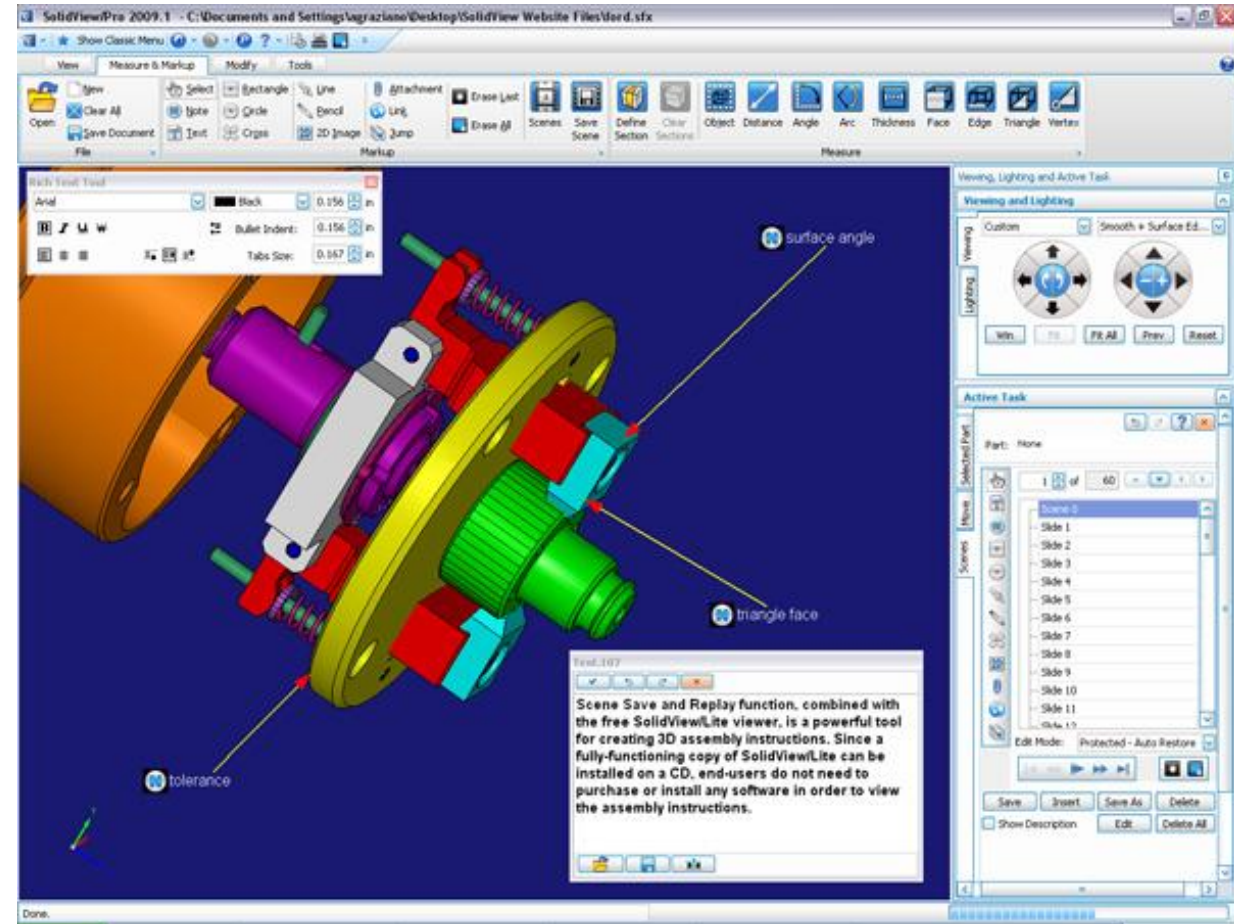
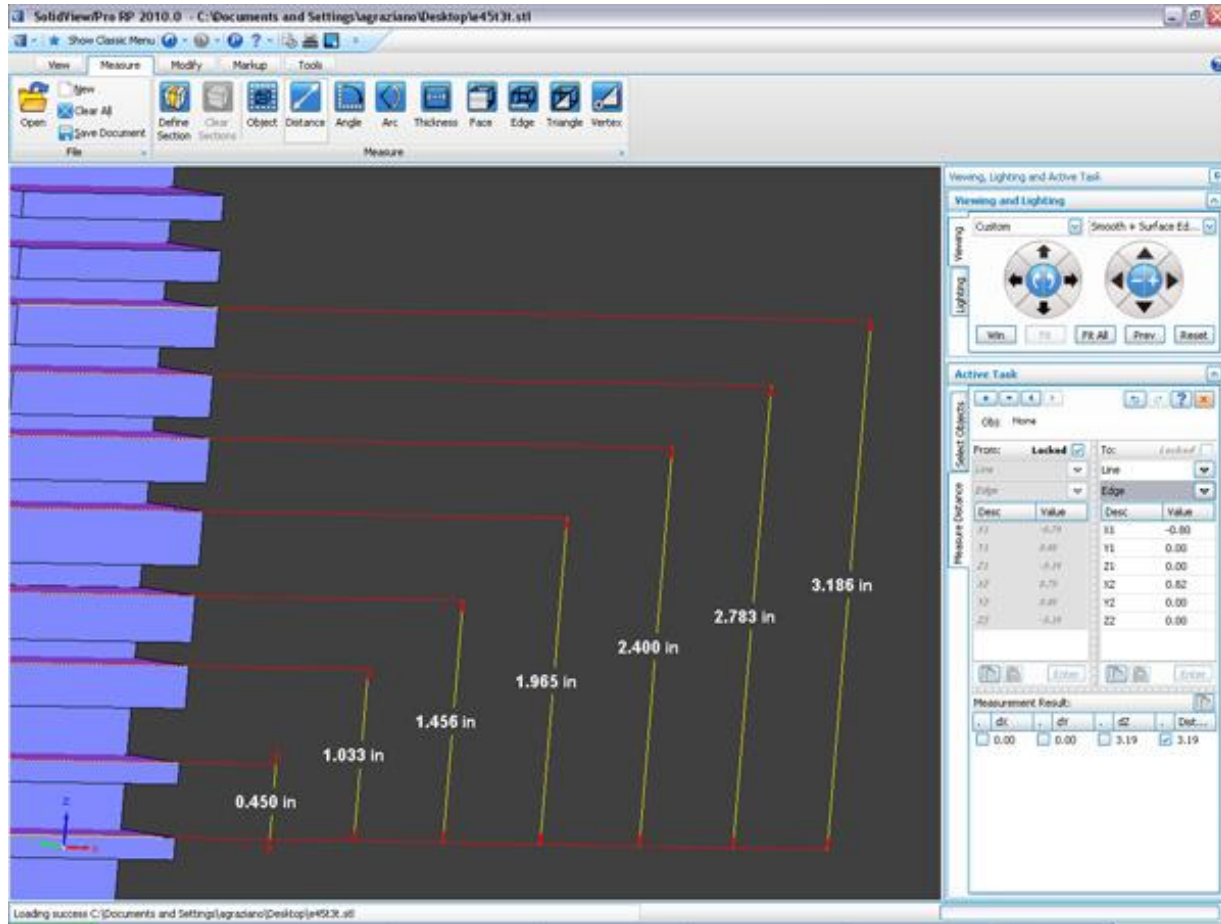
Design process:
Communication
with design team

It's a low-cost solution to access CAD data for

1. manufacturing engineers, 2. scientists, 3. structural engineers, 4. technical illustrators, 5. managers, 6. product managers and 7. sales people.



Solid view software environment



Advantages of solid view

- Solid-View software allows **non-CAD users** to easily view, measure and markup CAD data, opening up communication to all who need to be involved in the design process.
1. **Virtual reviews** of complex parts and assemblies
 2. Shop floor measurements from native CAD data
 3. Design collaboration across long distances, time zones and CAD systems
 4. Access to CAD data for technical illustrations, packaging, and marketing

MAGICS

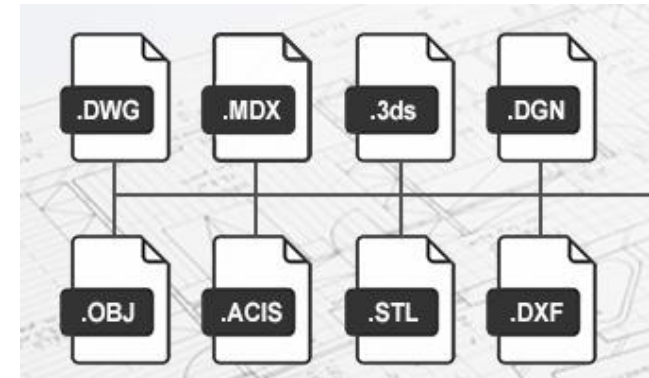
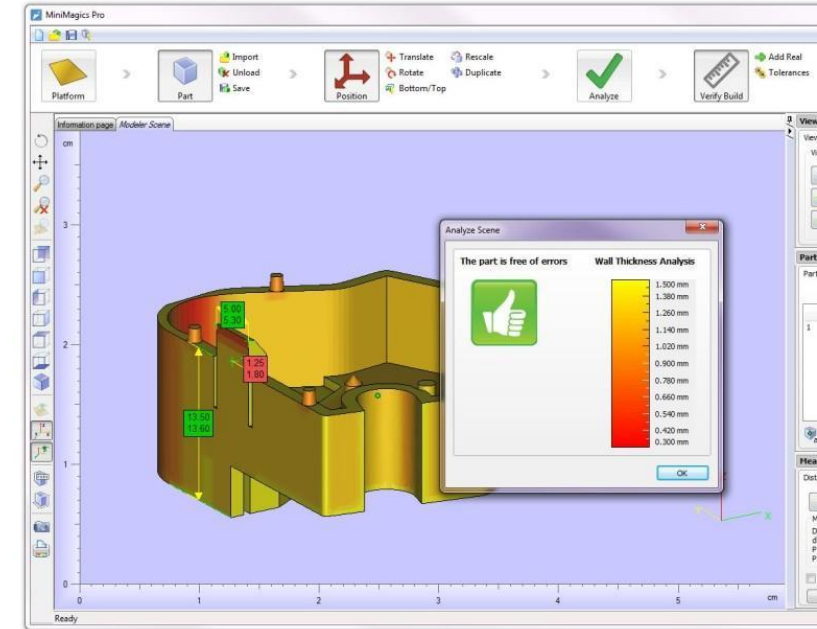
Magics is rapid prototyping software and is a key element of **Magics e-Solution Suit**.

Magics e-Solution Suit has full range software products.

It **streamlines**, **automates** and **boosts** almost every step in your rapid prototyping and manufacturing process.(RP&M).

MAGICS rapid prototyping software enables you to

1. Import a wide **variety of CAD formats**
2. Export STL files ready for rapid prototyping tooling and manufacturing.

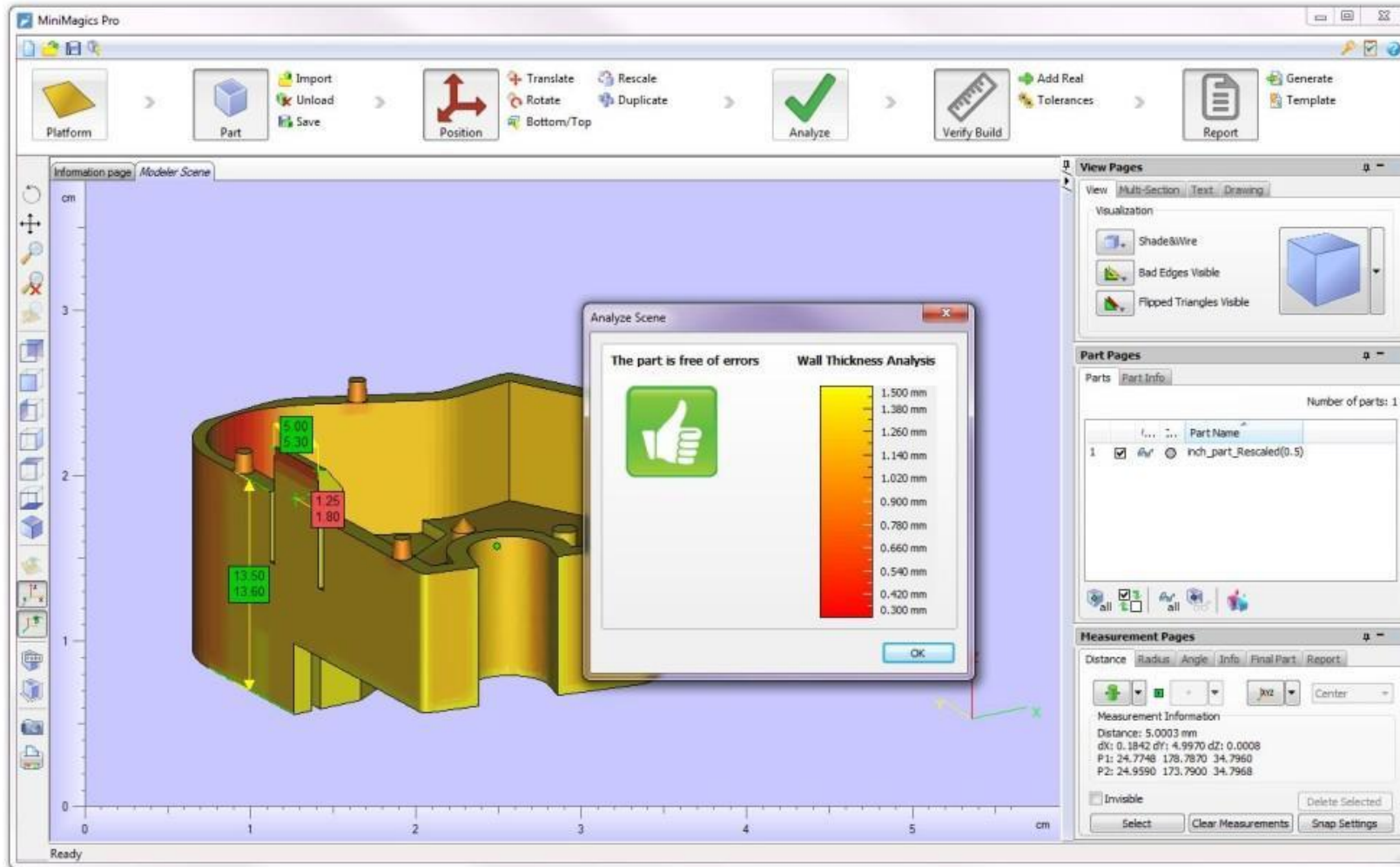


variety of CAD formats

Its application include

- A. repairing and optimizing 3D models;
- B. Analysing parts, making process related design changes on .stl files;
- C. Designing fixtures;
- D. Documenting projects;
- E. Production planning and much more.

MAGICS software environment



Key features of magics software

1. **Healing of STL files** into a solid mesh suitable for rapid prototyping

2. **Cutting STL models** into smaller sections when they are too big for the RP machine

3. **Creating offset wall section**, e.g. 1 mm wall on a solid bottle

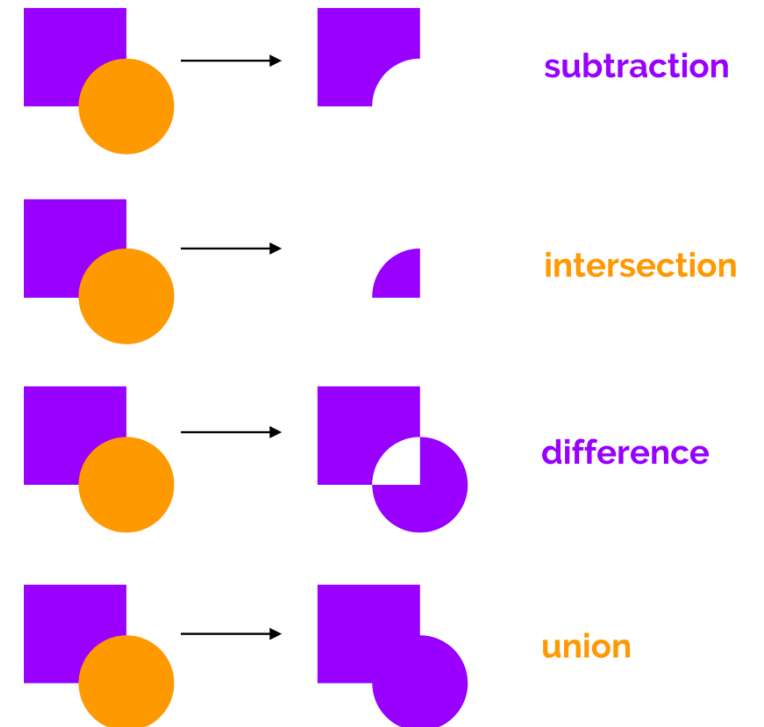


Key features of magics software

1. Shrink Wrap a new external STL skin onto a poor STL, very useful with architect models

2. Putting labels and logos onto models

3. Boolean operations on STL models



4. Magics communicator

- *1. Visualise*
- View STL, IGES, VDA and DXF 3 faces, with fast rotation, zooming and cross sectioning.
- *2. Annotate*
- Add 2D and 3D annotations, shapes, text and bitmaps.
- *3. Measure*
- Easily create 2D drawings from 3D files. Extensive feature recognition allows measuring of distances, radii and angles in 3D. Add tolerances and additional info.
- *4. Present*
- Make a 3D slide show with adjustable colours, shading and transparency.
- *5. Ease of use*
- Communicator's straight forward interface ensures that even non-CAD users will be comfortable with the program in no time.